

Munich Delivery Operations

Performance Analysis & Operational Forecasting

SLA Analysis | Order Forecasting | Driver Planning

4 Years

of Daily Operations Data

5

Business Questions Answered

3

Forecasting Models Compared

Python · Jupyter Notebook
Prophet · SARIMA · Holt-Winters
Matplotlib · Statsmodels

What This Analysis Covers

01

Data Foundation

4 years of Munich operations data : cleaning approach, quality checks, KPIs defined

02

Q1 · Part A

SLA & Service Quality

How often the 25-min target and service quality are met : year-over-year compliance trends

03

Q1 · Part B

Root Cause Analysis

What factors drive deviations from the 25-min target : which operational lever to pull first

04

Q2

Working Hours vs Fulfillment

Impact of courier working hours on the ability to serve potential orders : a counterintuitive finding

05

Q3

Order Volume Forecast

120-day forecast using Prophet, SARIMA and Holt-Winters : three models compared on same test set

06

Q4

Driver Planning : 90 Days

Required drivers per day using city efficiency (orders per working hour) : three planning scenarios

07

Q5

Assumptions & Methodology

Every assumption behind the driver estimation : stated transparently with justification

08

Additional

Revenue & SLA Simulation

€5.7M lost over 4 years : what a 2-min handling time reduction and better fulfillment would unlock

09

Recommendations

Three specific actions with quantified outcomes : ready for the logistics management team to action

Understanding the Dataset Before Touching Any Model

4 Years

Daily Data

Jan 2022 → Dec 2025

1,461

Daily Records

1,458 after cleaning

14

KPIs Analysed

17 cols · +16 derived

1 City

Munich Only

Germany operations

KEY METRICS ANALYSED

- **Total Time (mins)**
Full delivery time : order placed to customer handoff. Primary SLA metric.
- **Orders Delivered**
Successfully completed orders per day. Forecast target variable.
- **Potential (Demand)**
Total orders market wanted to place : including lost/unfulfilled orders.
- **Working Hours**
Total aggregate courier hours on shift across Munich per day.
- **Delivery Rating**
Average customer star rating (1-5) for the delivery service.
- **Fulfillment Rate** ★
 $\text{Orders Delivered} \div \text{Potential} \times 100$. Measures lost demand.
- **Orders per Working Hour** ★
City efficiency metric. Used directly in driver estimation formula.

★ Calculated metric : not in original dataset, derived during analysis

DATA QUALITY — ISSUES FOUND & ACTIONS TAKEN

Removed 3 rows

orders_delivered = 88,888

Potential on same days only ~5,500 : dummy value, unrecoverable.

Recalculated 3 rows

total_time_mins = 9,999

Rebuilt from assign_time + handling_time. Formula verified across all rows.

Recalculated 3 rows

order_distance_km = -9.90

Rebuilt from sub-distance components. Result 2.49-4.97 km : realistic.

Forward Filled 2 rows

daily_online_restaurants = 0

Active restaurants same days: 542 and 839 : clearly a recording error.

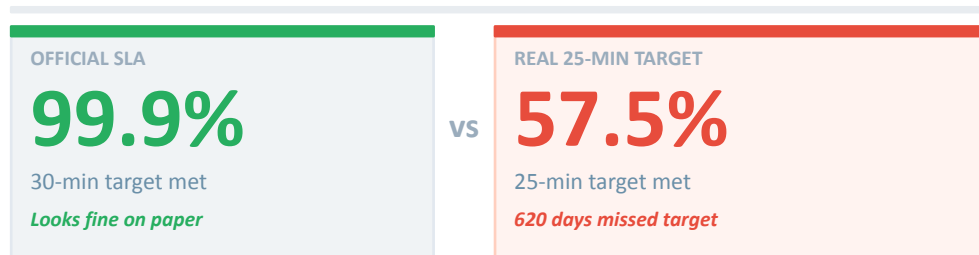
Interpolated 3 rows

delivery_rating : missing

Linear interpolation between neighbours. Rating is continuous : valid.

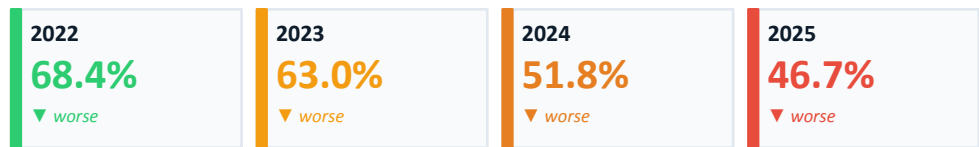
✓ Result: 1,461 → 1,458 rows · Zero missing values · Data quality confirmed

How Often Is the Delivery Target Actually Met?



⚡ Average delivery time = 24.71 mins — just 0.29 mins under the 25-min target. Any demand spike pushes Munich over the line.

YEAR-ON-YEAR DETERIORATION



✅ **Best: April (66.7%)**
Friday best day at 61.5%

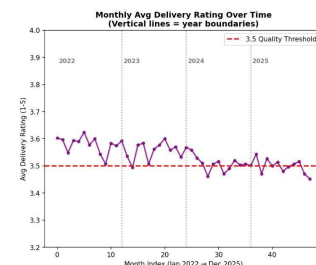
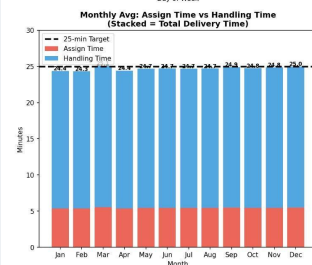
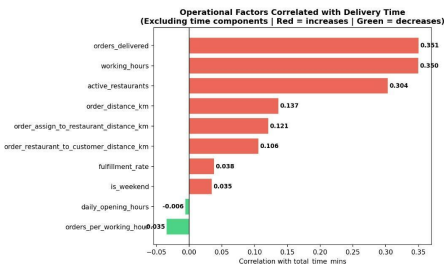
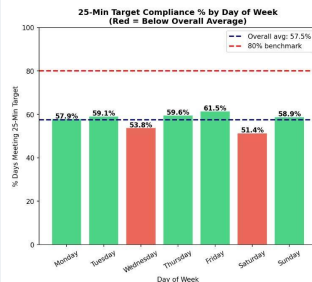
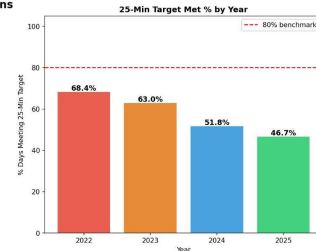
🔥 **Worst: March (47.6%)**
Saturday worst day at 51.4%

SERVICE QUALITY



ANALYTICAL EVIDENCE

Question 1 — Delivery Time & Service Quality Analysis Munich Operations



What Drives Deviations From the 25-Min Target?

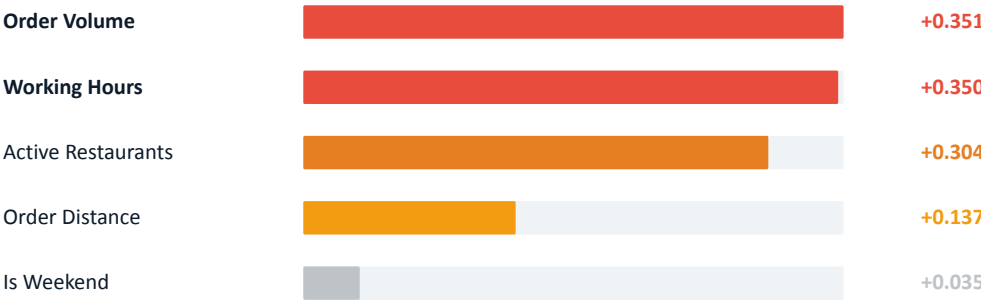
HOW DELIVERY TIME IS SPLIT



GAP BETWEEN SLA-MET vs SLA-MISSED DAYS



OTHER CONTRIBUTING FACTORS (correlation with total delivery time)



🎯 Reducing handling time by 2 mins → compliance 57.5% → 90.7% (+33 pct points)

ANALYTICAL EVIDENCE



More Couriers Does Not Mean Higher Fulfillment

THE COUNTERINTUITIVE FINDING

+0.027

Correlation

Working hours vs fulfillment rate

0.0008

R² Score

Working hours explains just 0.08% of fulfillment variation

FULFILLMENT RATE ACROSS WORKING HOUR BANDS



⚡ Range across ALL bands: only 0.30 percentage points — despite 75% more courier hours

WHAT ACTUALLY MATTERS & THE BUSINESS COST

REAL DRIVER

Potential Demand

Corr: -0.153

High demand days compress fulfillment regardless of couriers

BUSINESS COST

Lost Orders — +41%

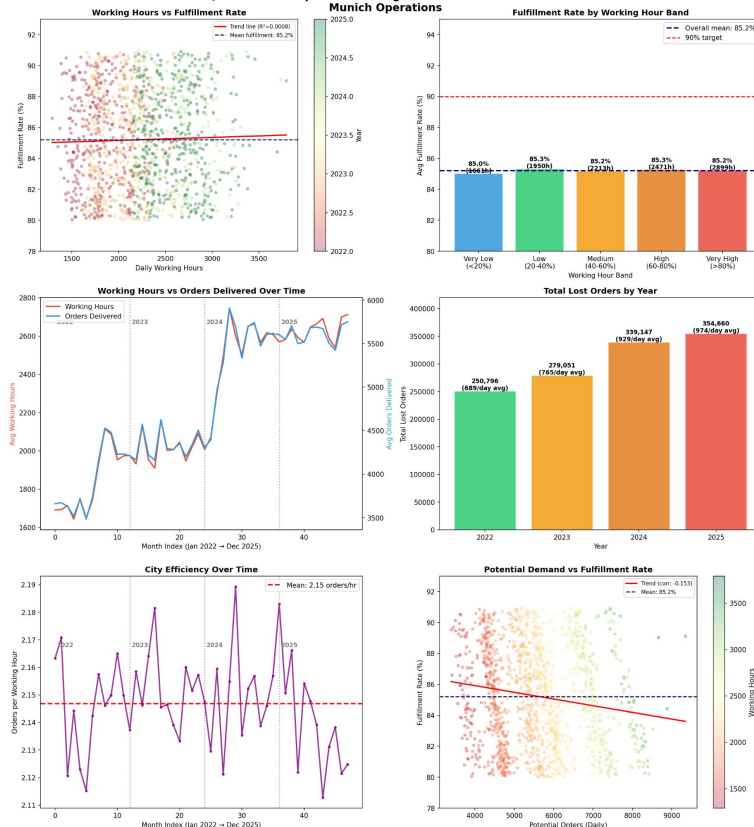
250,796 → 354,660

2022 → 2025. Even at stable 85% rate — market grows

✅ City efficiency flat at 2.15 orders/working hour for all 4 years — adding couriers has not improved productivity

ANALYTICAL EVIDENCE

Question 2 — Impact of Working Hours on Order Fulfillment



120-Day Order Forecast - Jan to Apr 2026

FORECASTING METHODOLOGY

ADF Test

p = 0.4422

Non-stationary
d=1 required

Trend Strength

0.929

Very strong
upward growth

Seasonal Strength

0.781

Strong weekly
pattern confirmed

THREE MODELS + ENSEMBLE TESTED · SAME 120-DAY TEST SET

Holt-Winters ETS 🏆

4.70%

✓ Excellent

Prophet

5.95%

✓ Good

Ensemble (avg)

5.66%

✓ Good

SARIMA(1,1,1)(1,1,1,7)

6.51%

✓ Good

FORECAST RESULTS (Holt-Winters · Jan-Apr 2026)

Jan 2026

5,528

orders/day

Feb 2026

5,501

orders/day

Mar 2026

5,487

orders/day

Apr 2026

5,475

orders/day

Pessimistic

4,937/day

Base ← Plan

5,498/day

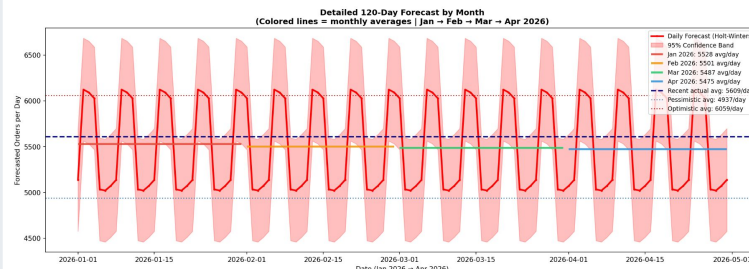
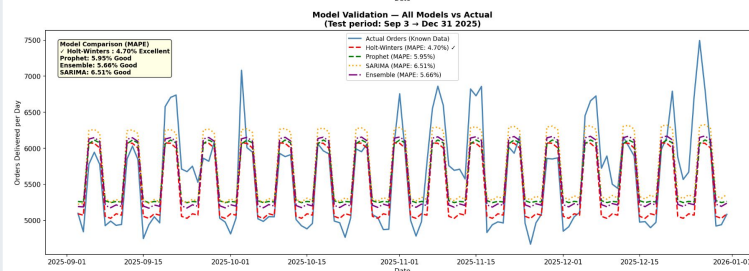
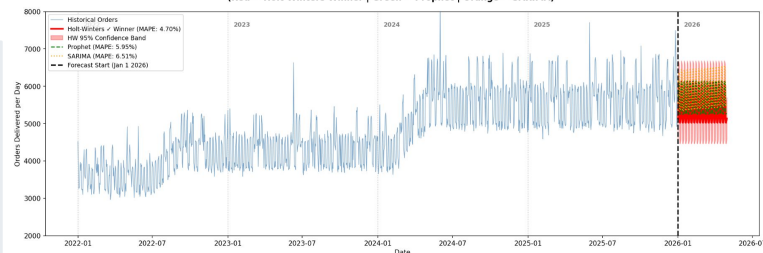
Optimistic

6,059/day

✓ Holt-Winters selected as planning model — MAPE 4.70% (Excellent) · Base forecast: 5,498 orders/day

ANALYTICAL EVIDENCE

Question 3 — Order volume forecast (120 days)
Munich Operations | Jan 2026 – Apr 2026
Four Models Compared: Prophet | SARIMA | Ensemble | Holt-Winters
Full Historical Data + 120-Day Forecast
(Red = Holt-Winters Winner | Green = Prophet | Orange = SARIMA)



Required Drivers : 90-Day Plan (Jan-Mar 2026)

ESTIMATION FORMULA

$$\text{Required Drivers} = \text{Forecasted Orders} \div (\text{City Efficiency} \times \text{Opening Hours})$$

Inputs: 5,506 orders/day · 2.147 orders/hr · 11.90 hrs weekday · 13.50 hrs weekend

PLANNING SCENARIOS (avg drivers/day · Jan-Mar 2026)

186 Pessimistic
Minimum floor — demand lower bound drivers/day

208 Base ← Recommended
Primary plan — Holt-Winters base forecast drivers/day

228 Buffered (+10%)
Safe plan — covers MAPE error + contingency drivers/day

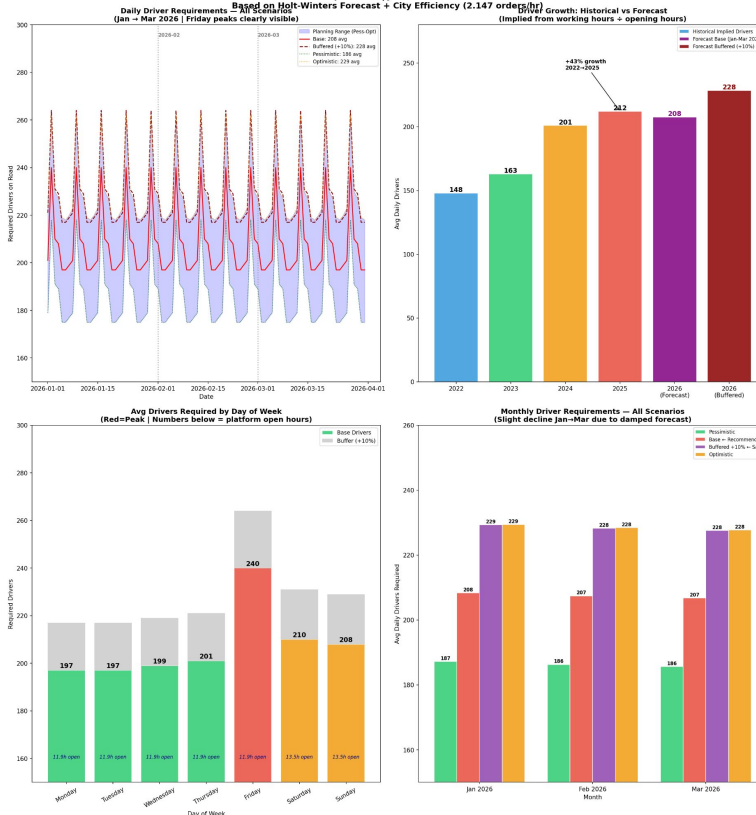
229 Optimistic
Upper bound — peak demand scenario drivers/day

DAY-OF-WEEK INSIGHT (base scenario)

Mon	Tue	Wed	Thu	Fri	Sat	Sun
197	197	199	201	240	210	208
11.9h	11.9h	11.9h	11.9h	11.9h	13.5h	13.5h

⚡ Friday needs MOST drivers (240) — shorter platform window (11.9h) vs Saturday (13.5h) concentrates demand

Question 4 — Required Drivers Estimation (90 Days)



Every Assumption Behind the Driver Estimation

Stated transparently - each assumption is justified by data, not guesswork

A1 City Efficiency = 2.147 orders/hr

Data validated

Mean across all 1,458 days: 2.147 orders/hr. Monthly average std deviation ± 0.02 across 48 months : stable enough to use as a fixed planning constant.

A2 Weekday opening hours = 11.90 hrs

Data validated

Calculated as mean of daily_opening_hours across Mon-Fri. Consistent across all years with negligible variance. Used in driver formula denominator.

A3 Weekend opening hours = 13.50 hrs

Data validated

Mean of daily_opening_hours for Sat–Sun. Longer window than weekdays : explains why Saturday needs fewer drivers than Friday despite more orders.

A4 Fulfillment rate remains at ~85%

Proven by Q2

Structural ceiling observed across all 4 years (80.0%-90.9%). Adding couriers did not improve it ($R^2=0.0008$). Demand growth is the binding constraint.

A5 Holt-Winters MAPE 4.70% holds forward

Test-set validated

Validated on 120-day held-out test set (Sep - Dec 31 2025). Seasonal patterns (weekly) and trend (damped) both confirmed stable over 4-year period.

A6 One driver works one full shift per day

Simplification

Simplifying assumption for planning purposes. Part-time shifts or split shifts would require operational data not available in this dataset. Flagged for refinement.

A7 No structural demand shocks in Q1 2026

Scenario covered

Holt-Winters damped trend naturally moderates extreme extrapolation. Pessimistic scenario (186/day) provides a lower-bound buffer if demand softens.

A8 +10% planning buffer = appropriate contingency

Industry standard

Buffer chosen to exceed model MAPE of 4.70% : covers forecast error plus operational contingency (illness, no-shows). Industry standard for logistics planning.

What Is the Financial Cost of Lost Orders - What Would Fixing It Unlock?

REVENUE OPPORTUNITY (AOV €26.00 · 18% commission)

€5.7M

4-Year Lost Commission

2022–2025 · 1,223,654 lost orders

€1.7M

Lost in 2025 Alone

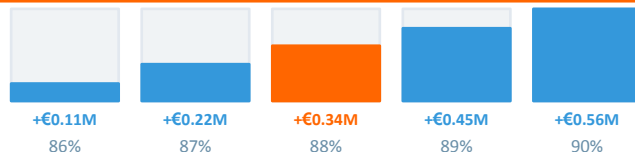
354,660 lost orders · 974/day avg

FULFILLMENT IMPROVEMENT VALUE

★ Recommended target: 85% → 88% = +€0.34M additional annual commission

€112K

per 1% improvement



SLA SIMULATION & RATING RETENTION LINK

SLA SIMULATION

-2 min handling time

57.5% → 90.7%

+33.2 pct points · 484 additional compliant days

Converts Munich to top performer

RATING & RETENTION LINK

SLA Met: 3.621 ★ vs Missed: 3.421 ★

Gap: 0.200 stars

0.200 star gap = ~1.0% repeat order rate uplift

Better SLA → Better ratings → Higher retention

Lost commission by year:

2022

€1.2M

2023

€1.3M

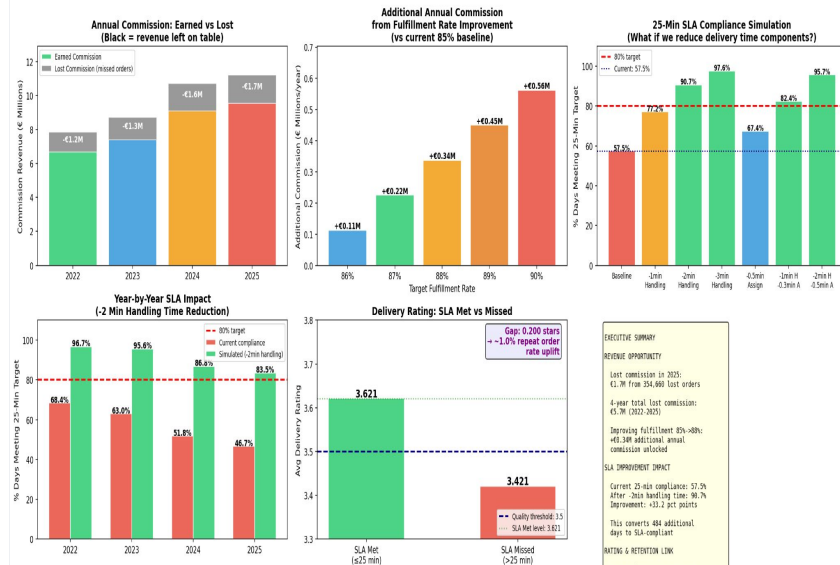
2024

€1.6M

2025

€1.7M

ANALYTICAL EVIDENCE

Additional Business Analysis
Lost Revenue Quantification + SLA Improvement Simulation
Munich Operations – Just Eat Takeaway.com

Three Actions : Each With a Quantified Outcome

Prioritised by impact · Each action is backed by data from this analysis

01

HIGHEST IMPACT

Reduce Order Handling Time by 2 Minutes

WHAT TO DO

Work with restaurant partners to streamline order preparation handoff. Target: reduce avg handling from 20.53 mins (missed days) to ≤18.50 mins.

EVIDENCE FROM ANALYSIS

Primary driver of SLA deviation - handling time gap = 2.19 mins between compliant and non-compliant days (78% of total delivery time).

QUANTIFIED OUTCOME

25-min compliance: 57.5% → 90.7% · +484 compliant days · Transforms Munich to top performer

02

REVENUE UNLOCK

Improve Fulfillment Rate from 85% to 88%

WHAT TO DO

Address demand capture during peak hours. Focus on Friday peak (240 drivers needed) and high-potential days where orders are being lost to capacity constraints.

EVIDENCE FROM ANALYSIS

85% fulfillment is a structural ceiling : not a staffing problem. Potential demand (corr -.153) is the real binding constraint. Each 1% recovered = €112K annual commission.

QUANTIFIED OUTCOME

+€0.34M additional annual commission · 85%→88% improvement · €112K per further 1%

03

OPERATIONAL

Reprioritise Friday Driver Scheduling

WHAT TO DO

Allocate 240 drivers on Fridays : currently the highest-demand day. Friday operates on an 11.9h window vs Saturday's 13.5h, meaning demand is more concentrated.

EVIDENCE FROM ANALYSIS

Friday requires 240 drivers/day : more than Saturday (210) despite weekend perception. Shorter platform window concentrates orders per hour. 208 base drivers needed Jan–Mar 2026.

QUANTIFIED OUTCOME

Eliminate Friday under-staffing · Support 5,528 orders/day (Jan 2026 forecast) · Reduce SLA breaches on peak day

Munich runs on 99.9% official SLA compliance : but only 57.5% of days actually meet the 25-min target.

€5.7M

Lost Commission

2022-2025 · 4 years

5,498

Orders/Day Forecast

Base · Jan-Apr 2026

208

Drivers Required

Daily · Jan-Mar 2026

90.7%

SLA With Fix

-2min handling time

THREE ACTIONS FOR THE LOGISTICS TEAM

01 Reduce order handling time by 2 minutes

→ Compliance 57.5% → 90.7% · +484 compliant days · Transforms Munich to top performer

02 Improve fulfillment rate from 85% to 88%

→ +€0.34M additional annual commission · Each further 1% = €112K/year

03 Reprioritise Friday driver scheduling : 240 drivers required

→ Shorter 11.9h window concentrates demand · More drivers than Saturday (210) despite perception

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April 2026

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